

GrowthOS — Integrations & API Setup Guide

Version: Post-audit (commit 570ca6b)

Setup by: _____

Date started: _____

Date completed: _____

Target environment: ☐ Production (https://growthos-blond.vercel.app) ☐ Local dev
(http://localhost:3000) ☐ Both

How to use: Work through each section **in order** — later integrations depend on earlier ones. Mark each step:

-  **Done** — configured and verified
-  **In progress** — started but not verified
-  **Skip** — not needed yet
-  **Blocked** — add note (App Review pending, billing, etc.)

After finishing a section, run the **Verify** steps at the bottom before moving on. When everything is wired, use [QA TESTING CHECKLIST.md](#) for full end-to-end validation.

Companion docs: [OPERATIONS.md](#) (live infra IDs) · [CODE AUDIT 2026-05-31.md](#) (known gaps)

Quick reference — what unlocks what

Integration	Env vars / setup	Unlocks in the app
Clerk	NEXT_PUBLIC_CLERK_*, CLERK_SECRET_KEY, CLERK_WEBHOOK_SECRET	Sign-up, sign-in, multi-tenant orgs
Supabase Postgres	DATABASE_URL	All data persistence (tenants, leads, posts, ...)
Vercel Blob	BLOB_READ_WRITE_TOKEN (auto on Vercel)	Media upload, AI images, voiceovers on production
Token encryption	TOKEN_ENCRYPTION_KEY	OAuth tokens + per-tenant API keys at rest
Anthropic	ANTHROPIC_API_KEY and/or /settings/api-keys	AI captions, copilot chat
OpenAI	OPENAI_API_KEY and/or /settings/api-keys	AI image generation in Post Studio
ElevenLabs	ELEVENLABS_API_KEY, ELEVENLABS_VOICE_ID and/or /settings/api-keys	AI voiceovers
Meta (FB + IG)	META_APP_*, webhooks, App Review	Connect IG/FB, publish, auto-replies
TikTok	TIKTOK_CLIENT_*, API approval	Connect TikTok, schedule/publish video
Canva	CANVA_CLIENT_*	"Edit in Canva" from Post Studio

Inngest	INNGEST_EVENT_KEY, INNGEST_SIGNING_KEY	Scheduled post publishing, lead email jobs
Resend	RESEND_API_KEY, RESEND_FROM	Email when a new lead arrives
Upstash Redis	UPSTASH_REDIS_*	Real rate limiting on lead form + webhooks (prod)

Section 0 — Prerequisites (before any API work)

#	Step	Details	Status	Notes
0.1	GrowthOS deployed and reachable	Production: https://growthos-blond.vercel.app		
0.2	GitHub repo cloned locally	~/Documents/growthos/		
0.3	apps/web/.env.local exists	Copy from apps/web/.env.local.example		
0.4	You can sign in to the app	Clerk auth working		
0.5	Onboarding completed for your test tenant	/get-started → finish all 5 steps		
0.6	Decide your public app URL	Production: https://growthos-blond.vercel.app · Local: http://localhost:3000		
0.7	Password manager / secrets doc ready	Never commit API keys to git		

Important: Every OAuth redirect URI and webhook URL below must use the **exact** value of `NEXT_PUBLIC_APP_URL` for that environment.

Section 1 — Core platform (required)

These are already live on production for GrowthOS. Use this section to verify or re-provision.

1A — Vercel (hosting)

#	Step	Details	Status	Notes
1.1	Open Vercel project	https://vercel.com/geoffrey-pearlmans-projects/growthos		
1.2	Confirm <code>main</code> auto-deploys	Push to GitHub → new deployment		
1.3	Confirm <code>VERCEL_FORCE_NO_BUILD_CACHE=1</code>	Production + Preview env vars (see <code>OPERATIONS.md</code>)		
1.4	Set <code>NEXT_PUBLIC_APP_URL</code>	https://growthos-blond.vercel.app (Production)		

Verify: Open production URL → landing page loads.

1B — Supabase Postgres (database)

#	Step	Details	Status
1.5	Open Supabase project	Ref <code>igizzkhcwednwbhqztgh</code> · https://supabase.com/dashboard/project/igizzkhcwednwbhqztgh	
1.6	Copy connection string (pooler)	Settings → Database → Connection string → Transaction pooler , port 6543	
1.7	Set DATABASE_URL on Vercel	Production + Preview scopes	
1.8	Set USE_LOCAL_DB=false on Vercel	Or omit — prod should never use PGlite	
1.9	Run catch-up migrations (if schema drift)	<code>DATABASE_URL='...' node scripts/apply-supabase-catchup.mjs</code>	
1.10	Local dev (optional)	Set DATABASE_URL in <code>.env.local</code> + <code>USE_LOCAL_DB=false</code> , or keep PGlite with <code>USE_LOCAL_DB=true</code>	

Verify: Sign in → complete onboarding → `/leads` loads without “schema out of date” error.

1C — Clerk (authentication)

#	Step	Details	Status	Notes
1.11	Open Clerk dashboard	https://dashboard.clerk.com		
1.12	Copy API keys	Publishable (<code>pk_test_...</code>) + Secret (<code>sk_test_...</code>)		
1.13	Set on Vercel	<code>NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY</code> , <code>CLERK_SECRET_KEY</code>		
1.14	Enable Google OAuth (optional)	Clerk → User & Authentication → Social connections → Google		
1.15	Enable Organizations	Clerk → Organizations (tenants map to Clerk orgs)		
1.16	Create webhook	Endpoint: <code>{NEXT_PUBLIC_APP_URL}/api/webhooks/clerk</code>		
1.17	Subscribe to events	<code>organization.created</code> , <code>organization.updated</code>		
1.18	Copy signing secret → <code>CLERK_WEBHOOK_SECRET</code>	Vercel + local <code>.env.local</code>		
1.19	Local <code>.env.local</code>	Paste same Clerk keys for local testing		

Note: `pk_live_` keys require a **custom domain** — `*.vercel.app` only works with `pk_test_` today.

Verify: Sign out → sign in with Google → land in app → nav links appear after onboarding.

1D — Token encryption (required in production)

#	Step	Details	Status	Notes
1.20	Generate a key	<code>openssl rand -hex 32</code>		
1.21	Set <code>TOKEN_ENCRYPTION_KEY</code>	Vercel Production + Preview + local <code>.env.local</code>		
1.22	Never rotate without a migration plan	Rotating invalidates stored OAuth tokens + tenant API keys		

Verify: App boots on Vercel without encryption errors; `/settings/api-keys` can save a key.

1E — Vercel Blob (media storage — required on Vercel)

#	Step	Details	Status	Notes
1.23	Open Blob store dashboard	Store <code>growthos-media</code> (<code>store_m90rm9zuMKzH5uS</code>)		
1.24	Confirm store linked to growthos project	Auto-provisions <code>BLOB_READ_WRITE_TOKEN</code>		
1.25	Confirm env var on Vercel	<code>BLOB_READ_WRITE_TOKEN</code> — Production, Preview, Development		
1.26	Local dev (optional)	Leave unset → files go to <code>.data/uploads/</code>		

Verify: `/create` → upload an image → appears in media library → survives page refresh.

1F — Upstash Redis (rate limiting — strongly recommended for production)

#	Step	Details	Status	Notes
1.27	Vercel dashboard → Storage → Create → Upstash Redis	Or https://upstash.com		
1.28	Link store to growthos project	Auto-populates env vars		
1.29	Confirm on Vercel	<code>UPSTASH_REDIS_REST_URL</code> , <code>UPSTASH_REDIS_REST_TOKEN</code>		
1.30	Redeploy after linking	Env vars picked up on next build		

Without this: lead-form rate limiting is in-memory only (weak on multi-instance Vercel).

Verify: Submit lead form 11x in one minute from same IP → 11th request rate-limited or friendly error.

Section 2 — AI providers (Post Studio + copilot)

Two ways to configure: (A) global env vars on Vercel / `.env.local` , or (B) per-tenant keys in **Settings → API Keys** (encrypted in DB). Tenant keys override env keys.

2A — Anthropic (captions, hooks, copilot) — recommended first

#	Step	Details	Status	Notes
2.1	Create Anthropic account	https://console.anthropic.com		
2.2	Add billing / credits	Required for API usage		
2.3	Create API key	Starts with sk-ant-		
2.4	Option A: Set ANTHROPIC_API_KEY on Vercel	All tenants use this key		
2.5	Option B: Add key in app	Settings → API Keys → Anthropic		
2.6	Local dev	Add to apps/web/.env.local OR use Settings → API Keys		

Verify: `/create` → "Generate copy with AI" → hook/caption/hashtags populate. Copilot chat (FAB or Create sidebar) streams a response.

2B — OpenAI (AI image generation)

#	Step	Details	Status	Notes
2.7	Create OpenAI account	https://platform.openai.com		
2.8	Add billing	Image API is paid		
2.9	Create API key	Starts with sk-		
2.10	Set OPENAI_API_KEY on Vercel and/or Settings → API Keys			

Verify: `/create` → Enhance tools → Generate image → image appears in media library.

2C — ElevenLabs (AI voiceovers)

#	Step	Details	Status	Notes
2.11	Create ElevenLabs account	https://elevenlabs.io		
2.12	Copy API key	Profile → API Key		
2.13	(Optional) Pick a voice ID	https://elevenlabs.io/voice-library → copy Voice ID		
2.14	Set ELEVENLABS_API_KEY (+ optional ELEVENLABS_VOICE_ID)	Vercel and/or Settings → API Keys		

Verify: `/create` → add caption text → Generate voiceover → audio asset in library.

Section 3 — Email notifications (Resend)

#	Step	Details	Status	Notes
3.1	Create Resend account	https://resend.com		
3.2	Add and verify sending domain	DNS: SPF, DKIM (optional DMARC)		
3.3	Create API key	Resend dashboard → API Keys		
3.4	Set <code>RESEND_API_KEY</code> on Vercel	Production + Preview		
3.5	Set <code>RESEND_FROM</code>	e.g. Growth05 <hello@yourdomain.com> — must use verified domain		
3.6	Redeploy			

Verify: Submit a lead on your public page → owner receives email (check spam). If no email, confirm Inngest is also wired (Section 4) — lead emails are triggered by background job.

Section 4 — Background jobs (Inngest)

Powers: ***scheduled post publishing*** (every 5 min cron) and ***new-lead email notifications***.

#	Step	Details	Status	Notes
4.1	Create Inngest account	https://www.inngest.com		
4.2	Create app named growthos (or sync from Vercel)	App ID in code: growthos		
4.3	Install Inngest Vercel integration (recommended)	Auto-syncs keys + registers <code>/api/inngest</code>		
4.4	Or set keys manually on Vercel	<code>INNGEST_EVENT_KEY</code> , <code>INNGEST_SIGNING_KEY</code>		
4.5	Confirm serve URL	<code>{NEXT_PUBLIC_APP_URL}/api/inngest</code>		
4.6	Redeploy production			
4.7	Inngest dashboard → Apps → sync	Functions should appear: <code>publish-scheduled-post</code> , <code>notify-new-lead</code>		
4.8	Local dev (optional)	Run <code>npx inngest-cli@latest dev</code> alongside <code>pnpm dev</code>		

Cron: `publish-scheduled-post` runs every **5 minutes** (`* / 5 * * * *`).

Verify: Schedule a post 6–10 minutes out with Meta connected → wait for cron → post status becomes **Published** (see QA checklist §7A.9).

Section 5 — Meta (Facebook + Instagram)

Hardest integration — requires a Meta Developer app, correct OAuth redirect, webhooks, and **App Review** for production permissions. Plan **1–3 weeks** for review.

5A — Meta business prerequisites (do this first)

#	Step	Details	Status	Notes
5.1	Facebook Page for your business	https://facebook.com/pages/create		
5.2	Instagram Business or Creator account	Not a personal IG account		
5.3	Link IG to Facebook Page	IG app → Settings → Account → Linked accounts → Facebook		
5.4	Confirm IG appears under Page	Meta Business Suite → Settings → Instagram		
5.5	Meta Business Portfolio (recommended)	https://business.facebook.com		

Without 5.1–5.4: OAuth may succeed but “No pages / Instagram accounts found.”

5B — Meta Developer App

#	Step	Details	Status	Notes
5.6	Create app at Meta for Developers	https://developers.facebook.com/apps/		
5.7	App type	Business (for Pages + Instagram)		
5.8	Add products	Facebook Login for Business, Instagram Graph API, Webhooks		
5.9	Copy App ID → META_APP_ID	Vercel + .env.local		
5.10	Copy App Secret → META_APP_SECRET	Vercel + .env.local — never expose client-side		
5.11	Generate webhook verify token	Random string, e.g. openssl rand -hex 16		
5.12	Set META_WEBHOOK_VERIFY_TOKEN	Same value in Vercel and Meta webhook config		
5.13	Add OAuth redirect URI	{NEXT_PUBLIC_APP_URL}/api/meta/oauth/callback		

5.14	Production example	<code>https://growthos-blond.vercel.app/api/meta/oauth/callback</code>		
5.15	Local dev example	<code>http://localhost:3000/api/meta/oauth/callback</code>		
5.16	Add app domain	<code>growthos-blond.vercel.app</code> (no <code>https://</code>)		
5.17	Add Privacy Policy URL	Required for App Review — can be a simple page		
5.18	Redeploy Vercel after env changes			

OAuth scopes GrowthOS requests (all need App Review for production):

```
pages_show_list
pages_read_engagement
pages_manage_metadata
instagram_basic
instagram_manage_comments
instagram_manage_messages
instagram_content_publish
```

5C — Meta Webhooks

#	Step	Details	Status	Notes
5.19	Meta app → Webhooks → Page object	Callback URL: <code>{NEXT_PUBLIC_APP_URL}/api/webhooks/meta</code>		
5.20	Verify token	Must match <code>META_WEBHOOK_VERIFY_TOKEN</code> exactly		
5.21	Subscribe to fields	<code>feed</code> (comments), <code>messages</code> (DMs)		
5.22	Subscribe app to your Facebook Page	Webhooks → Page → Subscribe		
5.23	Test webhook	Meta dashboard → Send test → expect 200		

Verify: Meta webhook GET challenge returns 200 with challenge string.

5D — Meta App Review (production)

#	Step	Details	Status	Notes
5.24	Switch app to Live mode	Only after review (or add test users in Dev mode)		
5.25	Submit permissions for review	Each scope listed in §5B		
5.26	Provide screencast + test credentials	Show OAuth connect, publish, auto-reply flow		

5.27	Add test users (dev mode only)	Meta app → Roles → Test users		
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Dev mode workaround: Add your Facebook account as Admin/Developer on the app — you can test without full review, but only for accounts with roles on the app.

5E — Connect in GrowthOS

#	Step	Details	Status	Notes
5.28	Go to Settings → Connections	/settings/connections		
5.29	Click Connect Instagram or Connect Facebook	Same Meta OAuth flow connects both		
5.30	Approve all requested permissions			
5.31	Confirm redirect back with success banner	"Meta accounts connected"		
5.32	Both cards show Connected	Instagram + Facebook if Page has linked IG		

Verify: Run QA checklist §7A (Meta connection + schedule + publish + auto-replies §8).

Section 6 — TikTok

*Requires TikTok Developer account and **Content Posting API** approval.*

#	Step	Details	Status	Notes
6.1	Register at TikTok for Developers	https://developers.tiktok.com		
6.2	Create an app	Enable Login Kit + Content Posting API		
6.3	Copy Client Key → TIKTOK_CLIENT_KEY	Vercel + .env.local		
6.4	Copy Client Secret → TIKTOK_CLIENT_SECRET			
6.5	Add redirect URI	{NEXT_PUBLIC_APP_URL}/api/tiktok/oauth/callback		
6.6	Production example	https://growthos-blond.vercel.app/api/tiktok/oauth/callback		
6.7	Submit for review (if required)	Content Posting often needs approval		
6.8	Redeploy Vercel			

6.9	Settings → Connections → Connect TikTok			
6.10	Approve permissions	Scopes: user.info.basic, video.publish		

Note: TikTok does **not** support comment/DM auto-replies in GrowthOS yet — publishing + copy-to-post only.

Verify: QA checklist §7B.

Section 7 — Canva Connect (optional)

#	Step	Details	Status	Notes
7.1	Canva Developer portal	https://www.canva.com/developers/		
7.2	Create Connect API integration			
7.3	Copy Client ID → CANVA_CLIENT_ID			
7.4	Copy Client Secret → CANVA_CLIENT_SECRET			
7.5	Add redirect URI	{NEXT_PUBLIC_APP_URL}/api/canva/oauth/callback		
7.6	Scopes (app requests)	design:content:read/write, asset:read/write		
7.7	Redeploy Vercel			
7.8	In Post Studio → Connect Canva / "Edit in Canva"	OAuth from /create or Canva flow		

Verify: QA checklist §7C.

Section 8 — Environment variable master checklist

Every var must be in `turbo.json#globalEnv` (already done in repo). Set on **Vercel** for production; copy to `apps/web/.env.local` for local.

Required (app won't work without these on Vercel)

Variable	Set?	Where	Notes
DATABASE_URL	☐	Vercel	Supabase pooler URL, port 6543
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY	☐	Vercel	pk_test_... for vercel.app

CLERK_SECRET_KEY	☐	Vercel	sk_test_...
CLERK_WEBHOOK_SECRET	☐	Vercel	From Clerk webhook
TOKEN_ENCRYPTION_KEY	☐	Vercel	openssl rand -hex 32
BLOB_READ_WRITE_TOKEN	☐	Vercel	Auto from Blob store link
NEXT_PUBLIC_APP_URL	☐	Vercel	https://growthos-blond.vercel.app

Strongly recommended (production)

Variable	Set?	Unlocks
UPSTASH_REDIS_REST_URL	☐	Distributed rate limiting
UPSTASH_REDIS_REST_TOKEN	☐	↑
VERCEL_FORCE_NO_BUILD_CACHE	☐	Already set — do not remove

AI (optional — or use Settings → API Keys)

Variable	Set?	Unlocks
ANTHROPIC_API_KEY	☐	AI copy + copilot
OPENAI_API_KEY	☐	AI images
ELEVENLABS_API_KEY	☐	Voiceovers
ELEVENLABS_VOICE_ID	☐	Voice selection

Social + design (optional)

Variable	Set?	Unlocks
META_APP_ID	☐	Facebook + Instagram
META_APP_SECRET	☐	↑
META_WEBHOOK_VERIFY_TOKEN	☐	IG/FB webhooks + auto-replies
TIKTOK_CLIENT_KEY	☐	TikTok
TIKTOK_CLIENT_SECRET	☐	↑
CANVA_CLIENT_ID	☐	Canva
CANVA_CLIENT_SECRET	☐	↑

Background + email (optional)

Variable	Set?	Unlocks
INNGEST_EVENT_KEY	☐	Scheduled publish + lead jobs

INNGEST_SIGNING_KEY	☐	↑
RESEND_API_KEY	☐	Lead notification emails
RESEND_FROM	☐	Sender address (verified domain)

Vercel env vars UI: <https://vercel.com/geoffrey-pearlmans-projects/growthos/settings/environment-variables>

After any Vercel env change: push a commit or trigger redeploy so the build picks up new values.

Section 9 — Recommended setup order (summary)

Use this if you want the fastest path to “full functionality”:

1. Core (Section 1) — already done on prod; verify
2. Anthropic (Section 2A) — biggest UX win, ~10 min
3. Upstash (Section 1F) — ~10 min, hardens prod
4. Inngest (Section 4) — unlocks scheduled publish + lead emails
5. Resend (Section 3) — pairs with Inngest for emails
6. OpenAI + ElevenLabs (2B–C) — optional AI extras
7. Meta (Section 5) — longest; start App Review early
8. TikTok (Section 6) — parallel with Meta review
9. Canva (Section 7) — optional polish

Parallel track: While Meta App Review is pending, you can complete Sections 1–4 and 2A–2C and test everything except live IG/FB publish and auto-replies.

Section 10 — URL cheat sheet (production)

Replace `{APP}` with your `NEXT_PUBLIC_APP_URL` (e.g. `https://growthos-blond.vercel.app`).

Purpose	URL
App home	<code>{APP}/</code>
Settings → Connections	<code>{APP}/settings/connections</code>
Settings → API Keys	<code>{APP}/settings/api-keys</code>
Meta OAuth callback	<code>{APP}/api/meta/oauth/callback</code>
Meta webhooks	<code>{APP}/api/webhooks/meta</code>
TikTok OAuth callback	<code>{APP}/api/tiktok/oauth/callback</code>
Canva OAuth callback	<code>{APP}/api/canva/oauth/callback</code>
Clerk webhook	<code>{APP}/api/webhooks/clerk</code>
Inngest serve	<code>{APP}/api/inngest</code>
Public lead page	<code>{APP}/p/{tenantSlug}/offer</code>

Section 11 — Troubleshooting

Symptom	Likely cause	Fix
"Clerk is not configured" banner	Env vars missing from build (<code>turbo.json#globalEnv</code>) or not set on Vercel	Re-check Vercel env vars; redeploy without stale cache
Media upload fails on production	<code>BLOB_READ_WRITE_TOKEN</code> missing	Link Blob store to Vercel project
"Meta is not configured" on Connections	<code>META_APP_ID</code> / <code>META_APP_SECRET</code> unset	Section 5B
OAuth redirect mismatch	Redirect URI in provider $\neq$ <code>{APP}/api/.../callback</code>	Must match exactly , including trailing slash rules
"No pages found" after Meta OAuth	IG not linked to Facebook Page	Section 5A
Webhook verify fails	<code>META_WEBHOOK_VERIFY_TOKEN</code> mismatch	Same string in Meta dashboard + Vercel
Scheduled posts never publish	Inngest not wired or Meta not connected	Sections 4 + 5
No lead emails	Resend or Inngest missing; domain not verified	Sections 3 + 4
AI features disabled	No Anthropic/OpenAI/ElevenLabs key	Section 2 or Settings → API Keys
Rate limit on lead form after testing	10 submits/min/IP	Wait 60s or test from different IP
TikTok OAuth 503	<code>TIKTOK_CLIENT_*</code> not set	Section 6

Section 12 — Final validation

When all sections above are , run the full [QA Testing Checklist](#) — especially:

- §6 — AI features
- §7 — Social connections
- §8 — Auto-replies (requires Meta + webhooks)
- §13 — End-to-end user journey

Export to PDF (optional, same as QA checklist):

```
# From repo root, if you have pandoc installed:  
pandoc docs/INTEGRATIONS_SETUP_GUIDE.md -o docs/INTEGRATIONS_SETUP_GUIDE.pdf
```

Or open the `.md` in Cursor / VS Code → Markdown PDF extension → Export.

Sign-off

Setup by	Date completed	Environment	Integrations live
		☐ Prod ☐ Local	☐ Core ☐ AI ☐ Meta ☐ TikTok ☐ Canva ☐ Inngest ☐ Resend

Generated from codebase at commit `570ca6b` . Update this guide when new integrations or env vars are added — see doc-currency rule in `CLAUDE.md` .